

A Multi-Source Dataset of Mwense Town Council Development Projects and Public Records

Group 33, CSC 4792 Data Mining and Warehousing

Authors: Musonda Katongo (2020583969), Charles Hangoma (2021414469), Nosiku Mukuka (2022030346), Aaron Simfukwe (2021376079), Clement Nkhoma (2020044153)

Affiliation: The university of Zambia, School of Natural and Applied Sciences

Department of Computing and Informatics

Abstract

This article describes a structured, source-traceable dataset of public development-project, budget, and planning records for Mwense Town Council, a local authority in Mwense District, Luapula Province, Zambia. The dataset consolidates information published across the council's website and four official PDF documents – the Integrated Development Plan 2024–2034, the 2024 and 2025 approved Constituency Development Fund (CDF) and capital project lists, and the 2022 CDF Guidelines – into a single 25-column table of 59 records (47 CDF projects, 9 capital projects, and 3 Integrated Development Plan/administrative records), together with a 6-row, 8-column source inventory and a 47-row CDF-only subset. Records were built with a documented, script-based pipeline (source download, PDF text extraction, manual verification of one scanned PDF, cleaning, and controlled-vocabulary normalisation) that preserves the original source wording per row (`raw_text`) and flags interpretive decisions (extraction notes). Fields absent from the source (for example ward, funding source, disbursed and spent amounts, dates, and beneficiaries) are left blank rather than estimated, and their extent is documented in the accompanying data dictionary and quality-control report. The dataset and its generation code are openly available on GitHub and Kaggle in pipe-separated CSV format, enabling reuse in studies of local-government transparency, CDF governance, public-finance reporting, and data-mining coursework requiring auditable, real-world administrative data with realistic missingness.

Specifications Table

Subject	Computer Science – Data Mining and Warehousing
Specific subject area	Curation of a traceable, structured dataset of Constituency Development Fund (CDF) and capital project approvals, Integrated Development Plan (IDP) content, and administrative records published by a Zambian local authority (Mwense Town Council)
Type of data	Tables (CSV, pipe-separated); a supplementary parsed-text CSV; a Jupyter notebook (.ipynb); Python scripts (.py); two PNG figures
Data collection	Records were compiled from the official Mwense Town Council website and four publicly posted PDF documents (the Integrated Development Plan 2024–2034, the 2024 approved CDF project list, the 2025 approved capital/CDF project list, and the 2022 CDF Guidelines). Source pages and PDFs were retrieved with Python (requests, BeautifulSoup) and text was extracted from PDFs with pypdf. One 2024 PDF is a scanned image; its

	project names and budget figures were manually transcribed from the document and cross-checked against the source wording, which is retained verbatim per row. No values were estimated or imputed; absent fields were left blank and explicitly flagged as missing.
Data source location	Institution: Mwense Town Council. Town/District: Mwense. Province: Luapula. Country: Zambia. Primary web source: https://www.mwensecouncil.gov.zm/
Data accessibility	Repository name: GitHub and Kaggle. Data identification number / link: https://github.com/venmor/csc4792-group33-mwense-town-council-dataset (code, notebook, and provenance files) and https://www.kaggle.com/datasets/charleshangoma/mwense-town-council-public-records-dataset-g-33 (dataset and starter notebook).
Related research article	None. This is an original data article prepared for the CSC 4792 Data Mining and Warehousing course assignment.

Value of the Data

- The dataset makes verifiable, source-linked evidence of Zambian local-government CDF and capital-project approvals available in a single structured table, reducing the effort needed to consolidate information currently scattered across a council website and several PDF documents.
- Researchers studying local-government transparency, decentralised public finance, or Constituency Development Fund governance in Zambia can reuse the dataset without repeating primary-source retrieval, since every record carries its `source_id`, `source_url`, `access_date`, and the original source wording (`raw_text`).
- The explicit and non-imputed treatment of missing values (for example the complete absence of disbursement, expenditure, and beneficiary data) makes the dataset a realistic teaching case for data-mining and data-warehousing courses that require students to design cleaning, quality-assurance, and missing-data-handling strategies for authentic, imperfect administrative data.
- The identical 25-column schema shared by the full-record table and the CDF-only subset supports direct comparison across constituencies (Mwense and Mambilima) and years (2024 and 2025), and can be extended by future data collection from the same council without restructuring.
- The fully scripted, reproducible acquisition-to-export pipeline (crawl, download, extract, clean, export) can be adapted to curate comparable open datasets for other Zambian town councils, supporting cross-council comparative research.

Data Description

The dataset is distributed as three pipe-separated (|) CSV files following the naming convention `db-unza26-csc4792-[description].csv`, together with the source-generating Jupyter notebook, four Python scripts, three intermediate parsed-text CSV files, and two PNG figures.

File (outputs/, pipe-separated)	Rows	Cols	Purpose
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db-unza26-csc4792-mwense-town-council-records.csv	59	25	Primary table: 47 CDF projects + 9 capital projects + 3 IDP/administrative records
db-unza26-csc4792-mwense-town-council-cdf-projects.csv	47	25	CDF-project-only subset, identical schema (no schema drift)
db-unza26-csc4792-mwense-town-council-sources.csv	6	8	Source inventory, SRC001–SRC006

The primary table (mwense-town-council-records.csv) contains one row per record and 25 columns that separate record content from provenance metadata. Each row is categorised as a `cdf_project`, `capital_project`, `idp_record`, or `administrative_record`. Of the 59 records, 47 are CDF projects approved in 2024 (26 rows: 13 in Mwense Central and 13 in Mambilima) or 2025 (21 rows: 10 in Mwense and 11 in Mambilima); 9 are capital projects approved in 2025, funded through the Basic Capital Grant, the Motor Licensing Fund, or locally generated revenue; and 3 are Integrated Development Plan and administrative records describing the council's ten-year development plan and its participatory planning structure. The full data dictionary is given below.

Field	Description	Type / unit	Missing-value meaning
record_id	Unique record identifier	String (REC001...)	Must not be missing
source_id	Link to the source inventory table	String (SRC001...)	Must be corrected if missing
source_url	URL where the evidence was obtained	String (URL)	URL unavailable
source_title	Human-readable source title	String	Title not recorded
source_type	website, webpage, or pdf	Categorical	Type not classified
publication_date	Source publication date	Date	Not stated in source
access_date	Date/time the source was accessed	Datetime (YYYY-MM-DD HH:MM)	Must not be missing
record_category	<code>cdf_project</code> , <code>capital_project</code> , <code>idp_record</code> , <code>administrative_record</code>	Categorical	Must not be missing
project_name	Project or record name	String	Not reported / not applicable
project_description	Detail given in the source	String	Not reported
sector	Normalised sector (Infrastructure, Health,	Categorical	Not reported

	Education, Water and sanitation, Roads, Transport, Energy, Security, Administration, Planning, Health and Education)		
ward	Ward, “All wards”, or “Not reported”	String	Source did not specify ward
community	Community or village	String	Not reported (100% of current rows)
constituency	Mwense, Mambilima, Mwense Central, or blank	String	Not applicable (e.g. capital/IDP rows)
institution	School, clinic, or market name	String	Not separately stated (100% of current rows)
funding_source	Reported funder	String	Not reported (84.7%; mostly CDF rows without an explicit funder line)
budget_amount_zmw	Reported budget or allocation	Numeric (ZMW)	Not reported (57.6%; all 2025 rows and 1 of 26 2024 rows)
amount_disbursed_zmw	Reported disbursed amount	Numeric (ZMW)	Not reported (100%; no source states it)
amount_spent_zmw	Reported spent amount	Numeric (ZMW)	Not reported (100%)
project_status	Reported status	Categorical (Approved)	Not reported
start_date	Start date	Date	Not reported (100%)
completion_date	Completion date	Date	Not reported (100%)
beneficiaries	Beneficiaries	String	Not reported (100%)
raw_text	Source wording, retained for verification	String	Extraction produced no text
extraction_notes	Interpretation or quality notes added during curation	String	No additional notes

The accompanying source inventory (mwense-town-council-sources.csv) records the six official sources from which every project and administrative record is drawn, each identified by a source_id referenced from the main table.

source_id	Type	Title	Topic
SRC001	website	Mwense Town Council official website and navigation pages	Council profile, departments, CDF, publications, and news
SRC002	webpage	List of approved projects page (?page_id=3026)	Links to the 2024 and 2025 project PDFs
SRC003	pdf	Integrated Development Plan 2024–2034	Planning, sectors, wards, committees, development priorities
SRC004	pdf	2024 approved CDF community projects	2024 Mwense Central and Mambilima CDF projects (scanned image; manually transcribed)
SRC005	pdf	2025 approved capital and CDF projects	2025 capital projects, CDF projects, wards, sectors, resolutions
SRC006	pdf	CDF Guidelines, February 2022	CDF governance, disbursement, utilisation, and accountability framework

Two supplementary, uncleaned intermediate files are retained in clean_data/ for transparency: raw_idp_unclean.csv (359 rows parsed from the Integrated Development Plan, used for contextual coverage of Ward Development Committees and is not merged into the main 59-row table because it lacks a per-ward membership list) and raw_projects_2025_unclean.csv (initial parse of the 2025 approved-project PDF prior to manual verification). Two figures are exported to figures/: records-by-sector-and-status.png (bar charts of record counts by sector and by status) and finance-ward-missingness.png (three panels covering budget availability by constituency, ward-reporting frequency, and missingness across the sparsest fields).

Experimental Design, Materials and Methods

Data collection and source acquisition

Data collection followed four stages implemented in a single Jupyter notebook (mwense-town-council-data-analysis.ipynb) that calls standalone, reusable Python scripts: (i) identify relevant pages and documents on the council website, (ii) extract text while retaining source metadata, (iii) clean and normalise records without discarding evidence, and (iv) analyse and export. A bounded, same-domain crawler (scripts/crawl_mwense_sources.py) starting from the council homepage and its assignment-relevant category pages produced a manifest of reachable pages, recording both successful and failed (HTTP 404) requests; twelve legacy content-* section URLs returned 404 and were excluded from further processing. A fixed-source downloader (scripts/download_mwense_sources.py) retrieved the four core PDF documents into raw_sources/, comparing local file size against the remote Content-Length header (HTTP HEAD) so that a top-to-bottom notebook re-run skips files that are already current; a --force flag forces refresh. Because the council's TLS certificate chain does not verify from the retrieval environment, requests were made with SSL verification disabled (verify=False) and are documented as such; this is a research-reproducibility accommodation rather than a data-content decision and does not affect record values.

Acquisition flags (RUN_SITE_CRAWL, RUN_SOURCE_DOWNLOAD, RUN_PDF_EXTRACTION) default to False in the delivered notebook so that a fresh run reuses the committed source files; setting them to True refetches from the live site.

Text was extracted from each downloaded PDF using pypdf (scripts/extract_mwense_pdfs.py), page by page, into raw_sources/extracted_text/, skipping PDFs whose extracted text is already newer than the source file. The extractor flags any output under 100 bytes as a likely scanned image. The 2024 approved-CDF-projects PDF triggered this flag (approximately 32 bytes of header text only); visual inspection confirmed it is a scanned image from which pypdf cannot recover tabular text. For this source only, project names and ZMW budget figures were manually transcribed directly from the PDF images and cross-checked twice against the document; the original source wording is preserved verbatim in the raw_text field, and an apparent amount inconsistency together with a heading/row-count mismatch in the Mambilima table (13 rows listed under a heading stating 12) are recorded in extraction_notes rather than silently corrected. All other sources (the Integrated Development Plan, the 2025 approved-project list, and the CDF Guidelines) yielded machine-extractable text.

Record construction and schema definition

A 25-field schema was defined before any record was populated, separating substantive fields (project_name, sector, ward, community, constituency, institution, funding_source, three monetary fields, project_status, three date/beneficiary fields) from provenance fields (record_id, source_id, source_url, source_title, source_type, publication_date, access_date, raw_text, extraction_notes). A single add_record() function enforced this schema for every row, pulling source_url, source_title, source_type, publication_date, and access_date automatically from the six-row source inventory by source_id, so that provenance could not be entered inconsistently. Records were added in four verified batches: 9 capital projects from the 2025 list, 10 CDF projects for Mwense Constituency (2025), 11 CDF projects for Mambilima Constituency (2025), 13 CDF projects for Mwense Central Constituency (2024), and 13 CDF projects for Mambilima Constituency (2024), followed by 3 non-project records (the Integrated Development Plan itself, its participatory planning structure, and the council profile) drawn from the Integrated Development Plan and the council website. Financial amounts were populated only where a source explicitly reported an allocation; the 2025 approved lists publish no monetary amounts, so budget_amount_zmw is missing for all 21 2025 CDF rows and all 9 2025 capital rows by design, not by omission.

Cleaning and quality assurance

Cleaning was performed on a working copy (cleaned_df) while the raw, as-constructed table (raw_df) was preserved unmodified for audit. Every text column was passed through a clean_text function that collapses internal whitespace with a regular expression and strips leading/trailing whitespace, converting empty strings to a null marker rather than leaving ambiguous blank text. Monetary columns (budget_amount_zmw, amount_disbursed_zmw, amount_spent_zmw) were passed through a clean_currency function that strips any non-numeric, non-decimal-point characters (currency symbols, thousands separators, stray text) before numeric coercion. Date columns (publication_date, access_date, start_date, completion_date) were parsed to datetime with invalid or absent values coerced to null rather than a default date.

Two controlled-vocabulary normalisations were applied and documented rather than left as free text: the sector label “Water” (used in the 2024 source) was mapped to “Water and sanitation” (the label used

consistently elsewhere), and CDF police-related entries recorded under “Home Affairs” were mapped to “Security” to match the equivalent 2025 Mwense “holdingcells” label. No sector, ward, funding-source, or status value was invented for a row that did not report one. Where a source did not report a ward, the field was set to the literal string “Not reported” (distinct from the legitimate value “All wards” used for district-wide projects) so that ward-coverage gaps are explicit and machine-detectable rather than silently blank.

A quality-control report was then generated over the cleaned table, verifying: (i) that all 59 record_id values are unique; (ii) that no row is missing a source_id or a record_category; (iii) the count of fully duplicated rows (0); and (iv) the count of negative values across the three monetary columns (0). A full missingness table (count and percentage per column) was produced and is summarised in the Limitations section and the data dictionary above. No row was deleted or estimated to reduce missingness; the quality-control step is diagnostic, not corrective, beyond the whitespace, currency-parsing, and controlled-vocabulary steps described above.

Analysis and export

Descriptive analysis (notebook Section 6) summarised the cleaned table by category, sector, and status; aggregated 2024 CDF budget amounts by constituency (the only year/category combination with materially complete monetary values); tabulated funding_source coverage; and cross-referenced the 359-row parsed Integrated Development Plan extract for Ward Development Committee and resolution context. Analysis was restricted to reporting what the sources state; absent information was treated as a reporting gap rather than as evidence that no such expenditure, disbursement, or ward assignment exists. Two static figures (bar charts and a three-panel missingness/coverage figure) were generated with matplotlib and seaborn and saved as PNG files.

The cleaned table, its 47-row CDF-only subset (identical 25-column schema, no schema drift), and the 6-row source inventory were exported as pipe-separated (|) CSV files named db-unza26-csc4792-[description].csv, per the course submission requirement. The notebook prints a compliance check confirming the pipe separator and filename prefix for each exported file at export time, and the full pipeline (installation check, acquisition, extraction, schema construction, cleaning, quality control, analysis, and export) executes top to bottom from a fresh clone without undisclosed manual steps, other than the documented manual transcription of the single scanned PDF described above.

Limitations

- Ward is recorded as “Not reported” for 38 of the 59 records because the underlying source did not specify a ward for those projects.
- The community, institution, beneficiaries, start_date, completion_date, amount_disbursed_zmw, and amount_spent_zmw fields are entirely absent (100% missing) across all 59 records because none of the six sources publish this information for any project.
- funding_source is missing for 50 of 59 records (84.7%), predominantly CDF rows that state no explicit funder line; budget_amount_zmw is missing for 34 of 59 records (57.6%), comprising all 30 rows drawn from the 2025 approved lists (which publish no monetary amounts) and 1 of the 26 2024 rows (an apparent amount anomaly in the source, flagged rather than corrected).

- All 56 project records carry the single status value “Approved”; the sources document approval decisions rather than implementation, so the dataset cannot be used to infer completion, delay, or expenditure outcomes.
- One source PDF (the 2024 approved-CDF-projects list) is a scanned image; its figures were transcribed manually rather than machine-extracted, and although cross-checked twice, transcription of scanned tabular data carries residual risk of transcription error, particularly for the flagged amount anomaly and the 13-row/12-heading discrepancy in the Mambilima table.
- Twelve legacy content-* section URLs referenced from the council's navigation structure returned HTTP 404 at the time of collection and could not be used as sources; some categories mentioned in the research questions (for example itemised Locally Generated Revenue Fund (LGEF) figures) are therefore not represented in the dataset because no reachable source publishes them.
- The dataset reflects a single access date per source and is not a longitudinal or continuously updated feed; repeated council publications (for example future annual approved-project lists) would need to be added through a repeat of the documented pipeline.

Ethics Statement

This work involved no human or animal subjects, no personal or sensitive data, and no experimentation. All records were compiled from documents and web pages that Mwense Town Council has published for public access. No individual beneficiary, household, or private citizen is identified in the dataset; project descriptions refer only to public infrastructure, institutions (schools, clinics), and named public facilities. The authors have complied with all applicable ethical guidelines for the collection and reporting of publicly available secondary administrative data.

CRedit Authorship Contribution Statement

Musonda Katongo: Conceptualization, Methodology, Writing – original draft. Charles Hangoma: Data curation, Software, Validation, Writing – review & editing. Nosiku Mukuka: Investigation, Data curation, Visualization. Aaron Simfukwe: Methodology, Formal analysis, Writing – review & editing. Clement Nkhoma: Investigation, Validation, Writing – review & editing. All authors jointly verified source provenance, reviewed the cleaned dataset, and approved the final manuscript.

Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

Data Availability

The dataset, generation notebook, and scripts are openly available at:

- GitHub repository (code, notebook, scripts, and provenance files): <https://github.com/venmor/csc4792-group33-mwense-town-council-dataset>
- Kaggle dataset (exported CSV files): <https://www.kaggle.com/datasets/charleshangoma/mwense-town-council-public-records-dataset-g-33>
- Kaggle starter notebook: <https://www.kaggle.com/code/charleshangoma/mwense-town-council-starter-notebook-group-33>

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